CS4641/CS7641 Homework 2 — Non-Programming

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Deadline: October 23rd, 11:59 pm ET

- No unapproved extension of the deadline is allowed. Refer to the syllabus for the late submission policy.
- Any section marked "Bonus for Undergrads" is optional extra credit for undergrads, but is **required for graduate students!** Any section marked "Bonus for All" is optional extra credit for everyone.
- Poorly formatted work may disrupt grading and may result in a **0 on affected questions**. See "Completing the Assignment" in the instructions below for more details.
- On most questions, work is required. Failure to show your work when expected may result in a **0 on affected questions**. See "Showing your Work" in the instructions below for more details.
- Failure to properly assign pages in Gradescope may disrupt grading and may result in a **0 on affected questions**. See "Submitting your Work" in the instructions below for more details.
- Discussion is encouraged! You are welcome to discuss at a whiteboard level with peers, but each student must write their own answers and explicitly mention any high-level collaborators. Plagiarism is a **serious offense**. You are not allowed to copy and paste, or paraphrase, or submit materials created or published by others, as if you created the materials. A good heuristic for cooperation is to discuss the questions together without discussing specifics, then write up your work and conclusions separately. If a conversation with a peer was meaningfully useful, please add their name to the Collaborators list below.
- Working with a generative-AI platform **may constitute plagiarism**. In line with the Joyner Heuristic being used by many classes at GT, you should treat collaboration with generative-AI as collaboration with a knowledgeable peer. It would be appropriate to ask about topics and formulae, but inappropriate to share the question and ask them to solve it. **Sharing the question or your work verbatim with an AI agent so as to generate an answer to the question is considered academic dishonesty and will be treated the same as any other incident of academic dishonesty.** If you're stuck, make a post on Ed, or visit office hours!
- Even using generative-AI for formatting your answers is **not permitted**. While we understand that LaTeX has a learning curve, you may not use any generative-AI platform to improve your writing or LaTeX formatting. These tools inherently produce output that may include a partial or complete solution to the question, including corrections to work you give it, even if purely prompted for syntactic use. Additionally, you have no custody over the data you give these platforms, which may leak or be used to inform subsequent training. Thus, while you are more than welcome to ask it about LaTeX commands and formatting tips, **sharing the question or your answer to a question verbatim with an AI agent, even purely for syntactic use, is considered academic dishonesty and will be treated the same as any other incident of academic dishonesty.** If you need formatting help, make a post on Ed, or visit office hours!
- All incidents of suspected dishonesty, plagiarism, or violations of the Georgia Tech Honor Code will be subject to the institute's Academic Integrity procedures. If we observe any (even small) issues detected by Gradescope or our TAs, **WE WILL DIRECTLY REPORT ALL CASES TO OSI**, which may, unfortunately, lead to a very harsh outcome, for example, academic probation or dismissal, a 0 grade for assignments concerned, and prohibition from withdrawing from the class.

Collaborators: *Please include whiteboard-level collaborators here, if any.*

Instructions

Completing the Assignment

- We provide you the source .tex files that contain the questions. We recommend completing the non-programming part of the assignment in a LaTeX editor. The ability to use LaTeX is a valuable skill in academia and in industry, and it's generally a clean way to render complicated math. Make sure that your work is formatted correctly. For example, you must submit $\sum_{i=0} x_i$ instead of sum_{i=0} x_i.
- You may use any LaTeX rendering tool you like! Overleaf is a great tool, with very few limitations. However, if you want finer control over packages and faster rendering, you may want to download a TeX distribution, a perl binary (if you're on Windows), and the "Latex Workshop" extension in VSCode to get live rendering in VSCode.
- If you're uncomfortable with using LaTeX, no worries! You can simply use the pdf version of the non-programming assignment and build your answers in any word or markdown processor you prefer. However, there are some caveats:
  1. **We will NOT grade handwritten work.** You must use some digital processor to render your work into a cleanly formatted pdf.
  2. It is **your responsibility** to ensure the questions are correctly numbered. While we will attempt to grade your work accurately, misnumbering may cause misinterpretation.
  3. It is **your responsibility** to ensure you're answering the question as it is written. We find that students who work without the questions embedded in their document tend to misinterpret the question more often. If you work in a separate document, read carefully and consider copying over the source question, even if its formatting may be lost.

Showing your Work

- When a question asks you to show your work, you must provide a justification for your solution. This may include calculations or text-based explanation. Though basic arithmetic may be combined, your work should be thorough and clear enough for a reader to follow your line of reasoning.
- When a question asks you to prove or provide a proof of some statement, you must provide a formal proof. These questions have a higher burden of work, and must constitute a valid inference. Again, while basic arithmetic can be combined, your work should be defensible as a formal proof. Any gaps in logic or unstated assumptions may result in deductions. Be careful using identities you may find online, since they may assume the fact you are trying to prove.
- If not otherwise indicated, no justification is needed, just the final answer.
- You will never be penalized for showing too much work, but you may be penalized for not showing enough.
- If you use an online resource, please cite it. We generally permit most resources, and so, citing it will only be to your benefit in the case of a plagiarism investigation (in the common case where multiple students utilize the same source).

Submitting your Work

- We will be using Gradescope for submission and grading of assignments. Complete the assignment, convert your assignment to a pdf, and submit it to the appropriate assignment denoted "Non-Programming" on Gradescope.
- When submitting your assignment on Gradescope, you are required to map pages of your pdf to subquestions on the homework. You should select a subquestion, then map it to **every** page which contains any portion of your solution to that subquestion.
- **You must map every page on which you have content that you would like be graded.** During the grading of each particular subquestion, we only see the pages you assign to that subquestion, so you must assign everything you would like us to grade.
- **Improperly mapped questions will receive penalties, up to and including 0 credit for the question.** You are permitted to submit a regrade request in this event; however, the review of such a request is at the sole discretion of the instructional staff and is not likely to be accepted.
- Some tips to correctly assign pages:

1. Underassigning pages is a common mistake. For example, if your answer to question 4.2 starts halfway down on page 16, goes through page 17, and ends with a single line bleeding into page 18, assign pages 16, 17, *and* 18 to question 4.2. Assigning just page 16 may incur a page assignment penalty, or the grader may just interpret your work as incomplete.
2. Keep in mind that overassigning is just as slowing to the grading process as underassigning is. Please try to keep your assignments tight around the content for the specific subquestion. E.g., assigning every page is just as useless as assigning no pages.

Again, assign every page that has content relevant to your answer that you would like for us to see while grading. If you don't assign it, we can't see it.

Point Distribution

Q1: Clustering Metrics [4pts]

Q2: EM Algorithm [15pts + 1% Bonus For All]

1. Performing an EM Update [15pts]
2. Gradient Ascent Comparison [1% Bonus For All]

Q3: Numeric Stability [6pts]

Q4: Alternate Implementation [5pts]

Q5: Semi-supervised Learning [2% Bonus For All]

Q6: K-Selection Problem [3% Bonus For All]

Points Totals:

- Non-Programming Base Points: 30pts
- Non-Programming Bonus for All: 6%
- For programming related points, see the Jupyter notebook.

1 Clustering Metrics [4pts]

In the Programming assignment (section 1.3), you should have implemented two clustering evaluation metrics: ARI and Silhouette Coefficient. Keeping those implementations in mind, answer the following questions.

a. For each metric, note the range of possible values, then describe what would constitute a "good" score and a "bad" score. [2pts]

Solution:

(a) ARI: Range is between -1 and 1.

1 is a good score indicating all pairs of datapoints have been tagged correctly,i.e, TP + TN = N and no incorrect clustering i.e, FP = FN = 0

-1 is a bad score indicating all pairs of datapoints have been tagged incorrectly,i.e, FP + FN = N and no correct clustering i.e, TP = TN = 0

0 is also a bad score indicating that the clustering is not better than random assignment of clusters

(b) Silhouette Coefficient: Range is between -1 and 1.

$$S(i) = \frac{b(i) - a(i)}{b(i)} = 1 - \frac{a(i)}{b(i)} \text{(Assuming b(i) ¿ a(i))}$$

$$S(i) = 1 \implies \frac{a(i)}{b(i)} = 0 \implies a(i) = 0 \implies \text{Average distance of points within a cluster is zero}$$

Therefore, Silhoette coefficient(SC) has the highest value when all points within each cluster are at the exact same point and the clusters are well separated (b(i) ¿ a(i)). Since this is improbable in a real dataset, a good score is when SC is approaching 1, because that means intra cluster distance is minimum and each cluster is distinctly placed.

$$S(i) = \frac{b(i) - a(i)}{a(i)} = \frac{b(i)}{a(i)} - 1 \text{(Assuming a(i) ¿ b(i))}$$

$$S(i) = -1 \implies \frac{b(i)}{a(i)} = 0 \implies b(i) = 0$$

$$\implies \text{i is misclassified into a wrong cluster, so it is far away from its actual cluster}$$

In reality, all datapoints don't get misclassified, hence whenever SC approaches -1, it is considered a bad score.

b. Identify the most notable difference between these two metrics and explain why this distinction is important when evaluating clustering results. [2pts]

ARI measures whether a pair of points lies in the same or different cluster and if that agrees with the ground truth, even if it is on a sample of all datapoints. For the $\binom{n}{2}$ pairs of data points, it measures how close the clustering is to the ground truth, adjusted for chance. However, this requires the ground truth to be available, so it is typically used to benchmark clustering algorithms on labeled datasets.

SC (Silhouette Coefficient), on the other hand, measures the geometry of the clusters formed in terms of cohesion and separation. It does not require any ground truth. It helps us evaluate clustering quality directly from the data, which is useful for deciding the optimal k value on unlabeled, real-world data

2 EM Algorithm [15pts + 1% Bonus For All]

2.1 Performing an EM Update [15pts]

Consider a univariate Gaussian Mixture Model (GMM) with two components. We'll add a slight variation: each component's mean and variation is parametrized together. In a normal GMM these are optimized separately, but this will save you some algebraic trouble. Our model is defined by the following parameter set $\theta = \{\alpha, \nu, \omega\}$, with:

$$\boldsymbol{z} = \begin{cases} 0 & \text{with } p = \alpha \\ 1 & \text{with } p = 1 - \alpha \end{cases}$$

$$p(x \mid \boldsymbol{z} = 0, \theta) = \mathcal{N}(x; \mu = 4\nu, \sigma^2 = 7\omega) = \frac{\exp\left(-\frac{(x-4\nu)^2}{14\omega}\right)}{\sqrt{14\pi\omega}}$$

$$p(x \mid \boldsymbol{z} = 1, \theta) = \mathcal{N}(x; \mu = 5\nu, \sigma^2 = 8\omega) = \frac{\exp\left(-\frac{(x-5\nu)^2}{16\omega}\right)}{\sqrt{16\pi\omega}}$$

Let's use this GMM to model the underlying distribution of a dataset $d = \{d^{\{1\}}, d^{\{2\}}, \cdots, d^{\{N\}}\}$. We aren't sure that it's a good fit yet, so we'll do an EM step to refine the parameters' (α, ν, ω) values.

a. Express the marginal probability density of the model $p(x \mid \theta)$ for an arbitrary value x. You may leave your answer in terms of $\mathcal{N}(x; \dots)$, no need to expand out Gaussians yet. Show your work. [2pts]

Hint

The rule of total probability may help.

Solution:

$$\begin{aligned} p(x \mid \theta) &= p(x \mid \boldsymbol{z} = 0, \theta)p(z = 0) + p(x \mid \boldsymbol{z} = 1, \theta)p(z = 1) \\ &= \mathcal{N}(x; \mu = 4\nu, \sigma^2 = 7\omega) * \alpha + \mathcal{N}(x; \mu = 5\nu, \sigma^2 = 8\omega) * (1 - \alpha) \end{aligned}$$

b. **E-Step:** Compute the posterior probability that a given arbitrary value x belongs to either component, the responsibilities of that value, $p(z = 0 \mid x, \theta)$ and $p(z = 1 \mid x, \theta)$. You may leave your answer in terms of $\mathcal{N}(x; \dots)$, no need to expand out Gaussians yet. Show your work. [4pts]

Hint

Bayes's rule may help.

Solution:

$$\begin{aligned} p(\boldsymbol{z} = 0 \mid x, \theta) &= \frac{p(x \mid \boldsymbol{z} = 0, \theta)p(z = 0)}{p(x \mid \theta)} \\ &= \frac{\mathcal{N}(x; \mu = 4\nu, \sigma^2 = 7\omega) * \alpha}{\mathcal{N}(x; \mu = 4\nu, \sigma^2 = 7\omega) * \alpha + \mathcal{N}(x; \mu = 5\nu, \sigma^2 = 8\omega) * (1 - \alpha)} \\ p(\boldsymbol{z} = 1 \mid x, \theta) &= \frac{p(x \mid \boldsymbol{z} = 1, \theta)p(z = 1)}{p(x \mid \theta)} \\ &= \frac{\mathcal{N}(x; \mu = 5\nu, \sigma^2 = 8\omega) * (1 - \alpha)}{\mathcal{N}(x; \mu = 4\nu, \sigma^2 = 7\omega) * \alpha + \mathcal{N}(x; \mu = 5\nu, \sigma^2 = 8\omega) * (1 - \alpha)} \end{aligned}$$

c. **M-Step:** To make an estimate on our parameters, we can use the maximum likelihood estimator. Making the standard assumption that data are i.i.d., likelihood can be expressed as:

$$\mathcal{L}(\theta; d) = p(d \mid \theta) = \prod_{n=1}^{N} p(d^{\{n\}} \mid \theta)$$

Unfortunately, this won't lend itself easily to optimization. Instead we'll maximize an evidence lower bound (ELBO) of the log-likelihood that is tight around the current values of the parameter set[1].

$$\ell(\theta; d) \geq \sum_{n=1}^{N} \sum_{k}^{\{0,1\}} \left[p(z = k \mid d^{\{n\}}, \theta_{\text{curr}}) \cdot \ln \left(p(d^{\{n\}}, z = k \mid \theta) \right) \right]$$

Hint

θ_{curr}, here, refers to the values of θ in the current iterative step. You may recognize the term $p(z = k \mid d^{\{n\}}, \theta_{\text{curr}})$ from lecture as responsibility—the probability that a point belongs in a particular cluster. This is what you computed in the E-Step, just applied to our dataset. Notationally, we can let $\tau_{nk} = p(z = k \mid d^{\{n\}}, \theta_{\text{curr}})$. For this step, all τ_{nk} are constants, whose derivative with respect to anything is 0.

From this expression, use the derivative to find the value of ω that maximizes the ELBO. Simplify your result as much as possible, and leave your answer in terms of τ, d, v, and N. Show your work. [9pts]

Solution:

Objective function here is the ELBO:

$$Q(\theta) = \sum_{n=1}^{N} \sum_{k}^{\{0,1\}} \left[p(z = k \mid d^{\{n\}}, \theta_{\text{curr}}) \cdot \ln \left(p(d^{\{n\}}, z = k \mid \theta) \right) \right]$$

Here,

$$p(z = k \mid d^{\{n\}}, \theta_{\text{curr}}) = \tau_{nk}$$

$$Q(\theta) = \sum_{n=1}^{N} \sum_{k}^{\{0,1\}} \tau_{nk} \cdot \ln \left(p(d^{\{n\}}, z = k \mid \theta) \right)$$

$$\begin{aligned} \ln(p(d^{\{n\}}, z = k \mid \theta) &= \ln(p(d^{\{n\}} \mid z = k, \theta) \cdot p(z = k \mid \theta)) \\ &= \ln(p(d^{\{n\}} \mid z = k, \theta)) + \ln(p(z = k \mid \theta)) \\ &= \ln(p(d^{\{n\}} \mid z = k, \theta)) + C \end{aligned}$$

(Here, $\ln(p(z = k \mid \theta)) = \alpha$ for k = 0 and $1 - \alpha$ for k = 1 So this can be considered a constant which will be removed during the derivative step.)

Now, let's evaluate

$$\ln(p(d^{\{n\}} \mid z = k, \theta))$$

for each k

(a) k = 0

$$\begin{aligned} \ln(p(d^{\{n\}} \mid z = 0, \theta)) &= \ln(\frac{1}{\sqrt{14\pi w}} \exp - \frac{(d^{\{n\}} - 4v)^2}{14w}) \\ &= \ln(\frac{1}{\sqrt{14\pi w}}) - \frac{(d^{\{n\}} - 4v)^2}{14w} \\ &= \frac{-1}{2} \ln 14\pi w - \frac{(d^{\{n\}} - 4v)^2}{14w} \end{aligned}$$

[1] A derivation of this ELBO can be found in the appendices of this document.

(b) k = 1

$$\begin{aligned}\ln(p(d^{\{n\}} \mid z=1,\theta)) &= \ln(\frac{1}{\sqrt{16\pi w}}\exp -\frac{(d^{\{n\}}-5v)^2}{16w}) \\ &= \ln(\frac{1}{\sqrt{16\pi w}}) - \frac{(d^{\{n\}}-5v)^2}{16w} \\ &= \frac{-1}{2}\ln 16\pi w - \frac{(d^{\{n\}}-5v)^2}{16w}\end{aligned}$$

$$\begin{aligned}Q(\theta) &= \sum_{n=1}^{N}\sum_{k}^{\{0,1\}} \tau_{nk}\cdot \ln\Big(p(d^{\{n\}}, z=k \mid \theta)\Big) \\ &= \sum_{n=1}^{N}(\tau_{n0}\cdot[\frac{-1}{2}\ln 14\pi w - \frac{(d^{\{n\}}-4v)^2}{14w}] + \tau_{n1}\cdot[\frac{-1}{2}\ln 16\pi w - \frac{(d^{\{n\}}-5v)^2}{16w}]) + C \\ &= \sum_{n=1}^{N} -\frac{\tau_{n0}}{2}(\ln 14\pi + \ln w) - \tau_{n0}\cdot\frac{(d^{\{n\}}-4v)^2}{14w} - \frac{\tau_{n1}}{2}(\ln 16\pi + \ln w) - \tau_{n1}\cdot\frac{(d^{\{n\}}-5v)^2}{16w} + C\end{aligned}$$

Taking derivative wrt w:

$$\frac{\partial}{\partial w}\frac{1}{w} = \frac{-1}{w^2} \text{ and } \frac{\partial}{\partial w}\ln w = \frac{1}{w}$$

$$\begin{aligned}&\frac{\partial Q(\theta)}{\partial w} = 0 \\ \implies &\sum_{n=1}^{N} -\frac{\tau_{n0}}{2w} + \tau_{n0}\cdot\frac{(d^{\{n\}}-4v)^2}{14w^2} - \frac{\tau_{n1}}{2w} + \tau_{n1}\cdot\frac{(d^{\{n\}}-5v)^2}{16w^2} = 0 \\ \implies &\sum_{n=1}^{N} -\frac{(\tau_{n0}+\tau_{n1})}{2w} + \tau_{n0}\cdot\frac{(d^{\{n\}}-4v)^2}{14w^2} + \tau_{n1}\cdot\frac{(d^{\{n\}}-5v)^2}{16w^2} = 0 \\ &(\tau_{n0}+\tau_{n1}=1) \\ \implies &\sum_{n=1}^{N} -\frac{1}{2w} + \tau_{n0}\cdot\frac{(d^{\{n\}}-4v)^2}{14w^2} + \tau_{n1}\cdot\frac{(d^{\{n\}}-5v)^2}{16w^2} = 0 \\ \implies &-\frac{N}{2w} + \frac{1}{w^2}\sum_{n=1}^{N}\tau_{n0}\cdot\frac{(d^{\{n\}}-4v)^2}{14} + \tau_{n1}\cdot\frac{(d^{\{n\}}-5v)^2}{16} = 0 \\ \implies &-Nw + 2\sum_{n=1}^{N}\tau_{n0}\cdot\frac{(d^{\{n\}}-4v)^2}{14} + \tau_{n1}\cdot\frac{(d^{\{n\}}-5v)^2}{16} = 0 \\ \implies &Nw = 2\sum_{n=1}^{N}\tau_{n0}\cdot\frac{(d^{\{n\}}-4v)^2}{14} + \tau_{n1}\cdot\frac{(d^{\{n\}}-5v)^2}{16} \\ \implies &w = \frac{1}{N}\sum_{n=1}^{N}\tau_{n0}\cdot\frac{(d^{\{n\}}-4v)^2}{7} + \tau_{n1}\cdot\frac{(d^{\{n\}}-5v)^2}{8}\end{aligned}$$

2.2 Gradient Ascent Comparison [1% Bonus For All]

EM cannot be applied to every problem, in that it only applies to a specific problem structure: one with latent variables. In the GMM problem, the hidden variable is the responsibilities. If we knew the ground truth responsibilities, we could simply use MLE to find the optimal model instantly. EM leverages this structure to get an efficient solution. However, there are other tools we could have used.

Gradient ascent/descent is a far more general optimization technique, in that it can work on any problem where you have a differentiable objective function (which is most problems in ML). To apply it to this problem, we could analytically compute the gradient of the log-likelihood with respect to the parameters, then take a small step in the parameter space in a direction that would increase the log-likelihood. Eventually, this process will find a maximum.

GA might be favorable here, not to mention that there's a veritable ocean of literature on improving gradient ascent/descent. It's not really leveraging specifics of the problem, but it's a prolific algorithm. We could just as easily solve this problem with gradient ascent, and maybe it would do it better.

$$\theta = \theta_{\text{curr}} + \eta \nabla \mathcal{L}(\theta_{\text{curr}})$$

What is one specific advantage that EM might have over gradient ascent for the problem explored in the previous section, that is to say, what is one reason you might choose EM over GA? Justify your answer with evidence. Either show your workings or cite from peer-reviewed literature. [1% Bonus For All]

3 Numeric Stability [6pts]

In the Programming assignment (section 2.3), you should have implemented the helper functions for GMM. The notebook contains some information which may be useful in answering these questions.

a. In the E-step of the GMM, we subtract the row maximum from the unnormalized model outputs before computing the responsibilities. Explain how numerical overflow or underflow could occur if this stabilization step were not performed, how subtracting the row maximum helps prevent these issues, and how numerical instability could affect the EM algorithm's convergence. Justify your answer (numerically, or brief explanation). [2pts]

Solution: In the E-step of the GMM, we calculate softmax of the result of log likelihood function. Softmax performs exponential function on the row elements which can explode the value resulting in NaN error. Since we need this responsibility value for our next M-step, this NaN error in E-step can affect M-step and stop the convergence of EM algorithm.

So we subtract the row max from each row element in the softmax function. This does not change the end result as shown below, but can avoid the NaN error effectively.

$$\begin{aligned}
\tau_{nk} &= \frac{\exp(logit_{i,j})}{\sum_{d=1}^{D} exp(logit_{i,d})} \\
&= \frac{\exp(logit_{i,j} - maxValue)}{\sum_{d=1}^{D} exp(logit_{i,d} - maxValue)} \\
&= \frac{\exp(logit_{i,j}) \cdot \exp -maxValue}{\sum_{d=1}^{D} exp(logit_{i,d}) \cdot \exp -maxValue} \\
&= \frac{\exp(logit_{i,j}) \cdot \exp -maxValue}{\exp -maxValue \cdot \sum_{d=1}^{D} exp(logit_{i,d})} \\
&= \frac{\exp(logit_{i,j})}{\sum_{d=1}^{D} exp(logit_{i,d})}
\end{aligned}$$

b. In the implementation instructions for logsumexp, you are told to subtract the row maximums, calculate logsumexp, then add them back. Prove (algebraically) how this process will return a correct result. More formally, let l be a row of D logits:

$$l = \{l_1, l_2, \cdots, l_D\}$$

Let s^* be the result of logsumexp applied directly to l:

$$s^* = \log\left(\sum_{i=1}^{D} \exp(l_i)\right)$$

Let s' be the result of logsumexp with our technique, subtracting the max before, adding the max after.

$$s' = \log\left(\sum_{i=1}^{D} \exp\left(l_i - \max_{j\in[1,D]}[l_j]\right)\right) + \max_{j\in[1,D]}[l_j]$$

You need to show that $s' = s^*$. [4pts]

Solution:

$$s' = \log\left(\sum_{i=1}^{D} \exp\left(l_i - \max_{j\in[1,D]}[l_j]\right)\right) + \max_{j\in[1,D]}[l_j]$$

Since

$$\max_{j\in[1,D]}[l_j]$$

is independent of i, we can call it

$$\alpha$$

for convenience

$$
\begin{aligned}
s' &= \log\left(\sum_{i=1}^{D}\exp\left(l_i - \max_{j\in[1,D]}[l_j]\right)\right) + \max_{j\in[1,D]}[l_j] \\
&= \log\left(\sum_{i=1}^{D}\exp\left(l_i - \alpha\right)\right) + \alpha \\
&\text{(Rewriting with } \exp\left(a+b\right) = \exp a * \exp b\text{)} \\
&= \log\left(\sum_{i=1}^{D}\exp(l_i)\cdot\exp\left(-\alpha\right)\right) + \alpha \\
&= \log\left(\exp\left(-\alpha\right)\cdot\sum_{i=1}^{D}\exp(l_i)\right) + \alpha \\
&\text{(Rewriting with } \log\left(a\cdot b\right) = \log a + \log b\text{)} \\
&= \log\left(\exp\left(-\alpha\right)\right) + \log\left(\sum_{i=1}^{D}\exp(l_i)\right) + \alpha \\
&= -\alpha + \log\left(\sum_{i=1}^{D}\exp(l_i)\right) + \alpha \\
&= \log\left(\sum_{i=1}^{D}\exp(l_i)\right) = s^*
\end{aligned}
$$

4 Alternate Implementation [5pts]

In the Programming assignment (section 2.5.1), you should have implemented a compression scheme for images using GMM clustering. The notebook contains some information which may be useful in answering this question.

Our compression scheme was dependent on clustering, but GMM isn't the only clustering solution. E.g., in section 1, you implemented K-means clustering. Consider how the clustering process would have been different if you had used K-means instead. Is there any way in which K-means might be more or less effective than GMM for this specific application? [5pts]
Solution:
In K-means, clustering takes place based on the Euclidean distance of datapoints from the nearest centroid. It tries to minimise the intra cluster distance using distorting as the objective function. GMM instead assigns probabilistic values to the datapoints and uses soft assignment. We finally take the mean of each component to assign the final cluster value to the pixels.
GMM will be more flexible than K-means in images since it can model clusters with varying shapes and spreads, while K-means works best with spherical clusters.
K-Means can be more effective than GMM in this application if the images result in clusters with spherical clusters. It is usually simpler and faster to use K-means in such cases. It is also significantly more space-efficient and faster since each cluster requires storing only a single RGB centroid vector (3 floats) plus the 2D assignment map. GMM requires storing the means, the mixture weights π_k, and full 3×3 covariance matrices Σ_k for every component, which increases metadata storage overhead and can reduce overall compression efficiency.
Therefore, neither method is guaranteed to produce a better image. The result depends on the image and can be compared using reconstruction error and visual quality for the same number of clusters.

5 Semi-supervised Learning [2.0% Bonus For All]

This section is accompanied by the semi-supervised data imputation part of the programming assignment (section 3.1).

1. What are some noticeable differences in the values produced by kNN imputation method, versus median value imputation? [0.5%]

2. We could have easily used the mean of each column to impute instead of the median. What advantage might median offer over mean for this use case? [0.5%]

3. Complete Case Analysis (CCA) is an alternative to imputation: simply drop every row that has a NaN. You only analyze the complete datapoints. This technique is not favored among statisticians. CCA is only justifiable on data that is Missing Completely At Random, i.e., the missingness itself is scatterred independently of the data. Why might Complete Case Analysis be unfavorable when the missingness is dependent on the values of the data itself? [1.0%]

6 K-Selection Problem [3% Bonus For All]

In the Programming assignment (section 1.1-1.3), you should have implemented the backbone of k-Means. The notebook contains some information which may be useful in answering this question.

In practice, the hardest part about using k-Means is picking an optimal value for k. Your assignment includes a data file "clusterable.csv", on which you'll attempt to do k-Means clustering. Read through the section informing you about how to evaluate clustering metrics in the notebook (section 1.4).

Your task is to estimate the optimal value of k for "clusterable.csv". To do this, you should pick any clustering metric, apply that clustering metric across a range of possible values for k, plot those results in a line plot (metric vs. k-value), embed the line plot into this document, and then provide an analysis of the graph and what k-value it suggests. Note that using only one metric may not be sufficient to support your answer. If you do this for real data in the real world, you'll want to use every tool available to you, and the tools will often disagree.

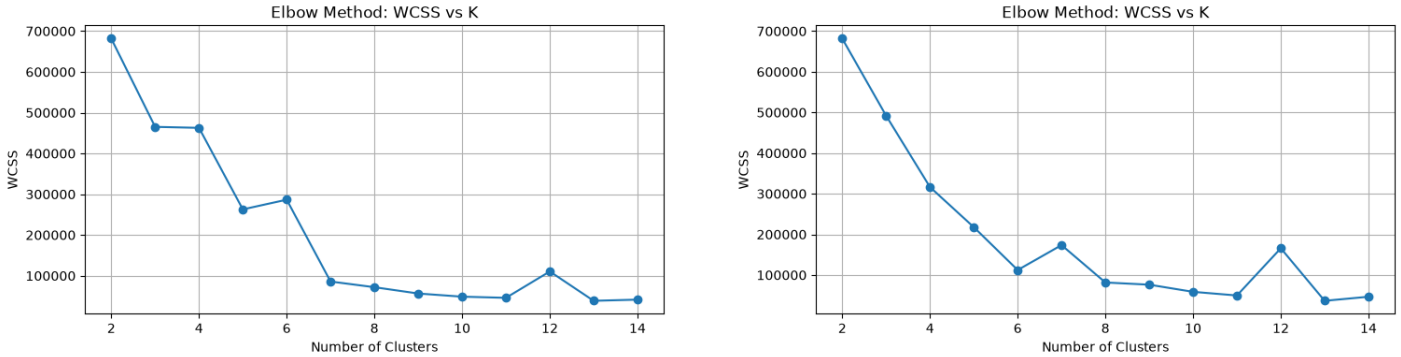
To receive credit, you should:

- For each metric:
  - Name the metric, briefly explain the metric, and mention what values would suggest a good k on that metric.
  - Embed a plot of the metric's performance vs. proposed k-values.
  - Analyze the plot. Make a guess for k based on the plot and briefly explain how confident this guess is.
  - Remember, negative results are good too! Even if your plot tells you nothing, it's still useful to a reader to see what you tried. You can simply ignore a useless result, but you should still include it.
- If you find your first metric to not be so convincing, you should repeat those steps for new metrics. Once you're satisfied with your answer, analyze all of the individual results and make a final recommendation based on some combination of the results.
- With your chosen value of k, cluster the data with that k-value one more time, make a scatter-plot with each cluster colored differently and embed that here. [3% Bonus For All]

Solution:

1. MCSS/inertia (loss function returned by the k-means function): It measures the sum of squared distances between each points and it is clustered center. In the best case, this value should be zero. But in a real dataset, any value approaching zero is a good value.

   Different initialisations produced different plots of metric vs proposed k-values. Since distortion is generally expected to decrease as k increases, I picked the below plot. The rejected plots are given in later.



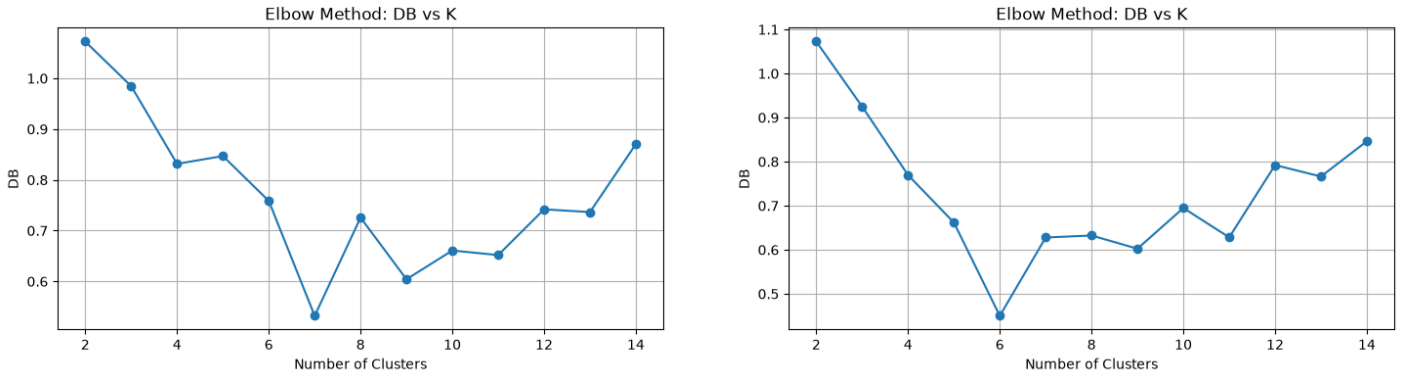
(a) Initialization 1

(b) Initialization 2

Figure 1: WCSS metric vs. Proposed K values

   In the first initialization [Figure 1a] , there is a visible elbow at k=7 after which the distortion varies much more slowly. In the second initialization [Figure 1b], there is a visible elbow at k=8, after which distortion varies only once at 12. This iteration was picked because the corresponding result in metric 2 Davies–Bouldin (DB) created a distinct result at k=6. Using this metric I would choose k=7. However, I would refer to Davies–Bouldin (DB) index before making the final decision.

2. Davies–Bouldin Index (DB) measures the ratio of cohesion to separation for each cluster. A score close to 0 means the clusters are tight but distinct from other clusters. I chose this over Silhoette Coefficient since SC timed out as it needs to be calculated for each datapoint.



(a) Initialization 1

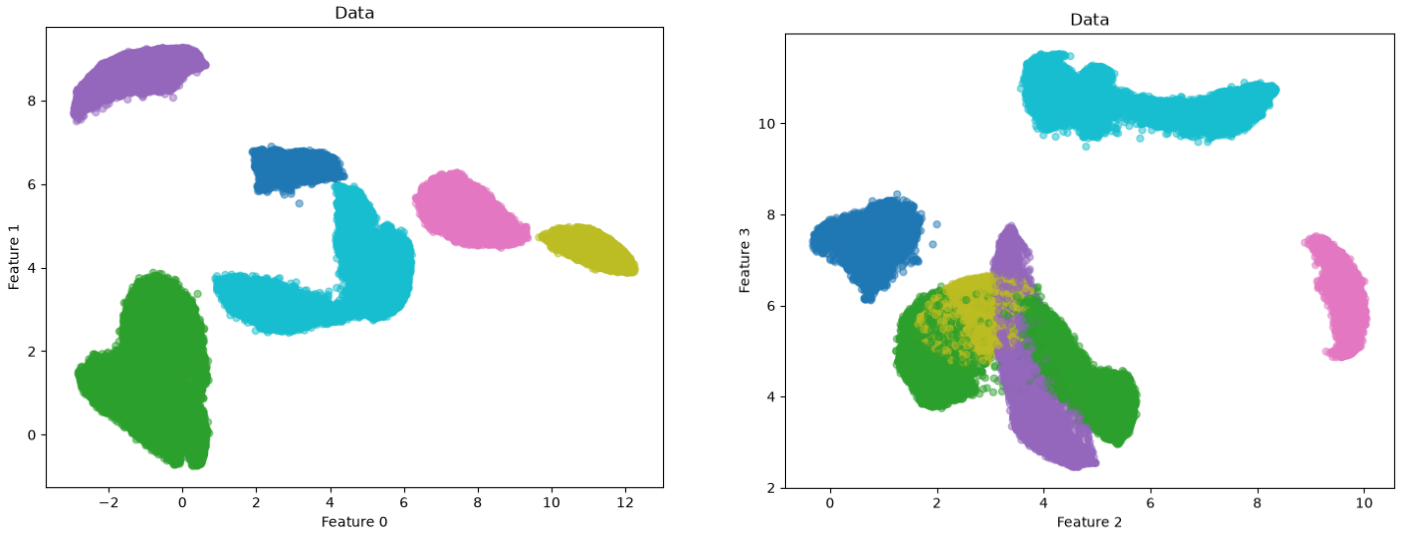
(b) Initialization 2

Figure 2: DB Index vs. Proposed K values

In the first initialization [Figure 2a], k=7 has the lowest value. In the second initialization [Figure 2b], k=6 produces a value less than 0.5, lesser than k=7.

Considering k values of 6 and 7 were identified from both the methods, k = 6 has a lower value in DB-index, I chose 6 as the final score.

Here is the scatter plot it generated for the various combinations of dimensions:



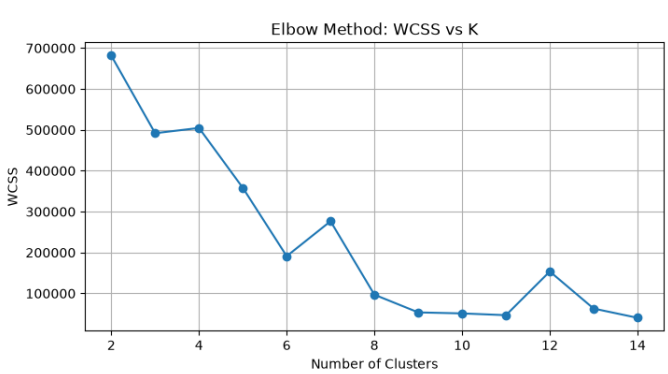
(a) Feature 0 and Feature 1 2D plot

(b) Feature 2 and Feature 3 2D plot

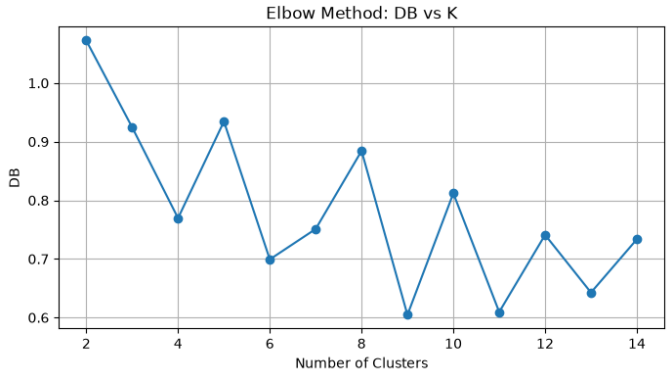
Figure 3: Scatter plot with k=6

For the different 2D scatter plots of feature combinations, it was able to create distinct clusters, although in some combinations, there were overlapping clusters.

Rejected plots: Here the distortion values changed multiple times in the graph and didn't produce a clear result. Hence I didn't choose them.



(a) Rejected Initialization for WCSS



(b) Rejected Initialization for DB

Figure 4: WCSS metric vs. Proposed K values

Appendix

The appendix will include any information that is extraneous to the questions asked, but still informative. This content is not necessary for the homework, only if you're curious about further math.

Justification for the ELBO

In the implementation of GMM, we maximize a tight lower bound, so-called, Evidence Lower BOund (ELBO). This is because the traditional MLE approach isn't very approachable—maybe numerically, but not analytically.

$$\begin{aligned}
\mathcal{L}(\theta; d) &= \prod_{n=1}^{N} p(d^{\{n\}} \mid \theta) \\
&= \prod_{n=1}^{N} \sum_{k=1}^{K} p(d^{\{n\}}, z_k \mid \theta) \\
\ell(\theta; d) &= \ln \left[\prod_{n=1}^{N} \sum_{k=1}^{K} p(d^{\{n\}}, z_k \mid \theta) \right] \\
&= \sum_{n=1}^{N} \ln \left[\sum_{k=1}^{K} p(d^{\{n\}}, z_k \mid \theta) \right]
\end{aligned}$$

Log of a sum won't give a clean gradient to zero out, but Jensen's might help us commute the sum through the log. That probability can be expanded (since $d^{\{n\}}$ and z_k are conditionally independent), but even if we got something more Jensen's like, the weights would still be conditioned on θ. We can't use variable weights, even if they are guaranteed to sum to one, they need to be constants for Jensen's (θ is a variable of the function). We can introduce a term dependent on the current parameters θ_{curr} (that sums to 1 over K), and maybe make some guarantees about the update range of the parameters.

$$\begin{aligned}
\ell(\theta; d) &= \sum_{n}^{N} \ln \left[\sum_{k}^{K} \left(\frac{p(z_k \mid d^{\{n\}}, \theta_{curr})}{p(z_k \mid d^{\{n\}}, \theta_{curr})} \right) \cdot p(d^{\{n\}}, z_k \mid \theta) \right] \\
&= \sum_{n}^{N} \ln \left(\sum_{k}^{K} p(z_k \mid d^{\{n\}}, \theta_{curr}) \cdot \frac{p(d^{\{n\}}, z_k \mid \theta)}{p(z_k \mid d^{\{n\}}, \theta_{curr})} \right) \\
&\quad \text{via total probability, } \textstyle\sum_k^K p(z_k \mid \cdots) = 1 \\
&\quad \text{via second derivative test, ln is concave} \\
&\quad \text{thus, } \ln(\mathbb{E}[x]) \geq \mathbb{E}[\ln(x)] \\
\therefore \ell(\theta; d) &\geq \sum_{n}^{N} \sum_{k}^{K} p(z_k \mid d^{\{n\}}, \theta_{curr}) \cdot \ln \left(\frac{p(d^{\{n\}}, z_k \mid \theta)}{p(z_k \mid d^{\{n\}}, \theta_{curr})} \right)
\end{aligned}$$

$$= \sum_{n}^{N} \sum_{k}^{K} p(z_k \mid d^{\{n\}}, \theta_{curr}) \cdot \left(\ln p(d^{\{n\}}, z_k \mid \theta) - \ln p(z_k \mid d^{\{n\}}, \theta_{curr}) \right)$$

We can further relax this lower bound by simply dropping the second log term. Since $p(z_k|...)$ is a probability and we've already implicitly assumed that it's never 0 or 1 in our E-step, which would mean that one of the components is useless (which is not possible for a Gaussian anyways), then we know $p(z_k \mid d^{\{n\}}, \theta_{curr}) \in (0, 1)$. Thus, $\ln\ p(z_k \mid d^{\{n\}}, \theta_{curr}) \leq 0$. Since this term is always negative, and it's currently being subtracted, dropping this term would strictly decrease the value, which is just a loosening of the lower bound.

$$\ell(\theta; d) \geq \sum_{n}^{N} \sum_{k}^{K} p(z_k \mid d^{\{n\}}, \theta_{curr}) \cdot \ln \left(p(d^{\{n\}}, z_k \mid \theta) \right)$$

definition of E-step's tau, verbatim

$$\ell(\theta; d) \geq \sum_{n}^{N} \sum_{k}^{K} \tau_{nk} \cdot \ln \left(p(d^{\{n\}}, z_k \mid \theta) \right)$$

for notation's sake, call it

$$\ell(\theta; d) \geq Q(\theta, \theta_{\text{curr}})$$

And so, we get this function, Q, which is a valid lower bound on the log-likelihood ($\ell(\theta; d) \geq Q(\theta, \theta_{\text{curr}})$), it's evidenced by the E-step's estimate, and also, it should be tight if you don't go anywhere ($\ell(\theta_{\text{curr}}; d) = Q(\theta_{\text{curr}}, \theta_{\text{curr}})$).

This is all well and good, but how can we be sure this iterative process will actually work? Honestly the math is above my head quite a bit, but I was able to prove that the optimal value for θ is shared, so if the current parameters constitute a locally maximum likelihood for the GMM, the EM process will stay stationary.

We can prove that with contradiction. To begin, assume the opposite of the implication ($p \wedge \neg q$):

- "the current parameters constitute a locally maximum likelihood for the GMM" means that θ_{curr} is a local maximum for $\ell(\theta; d)$
- "the EM process will **NOT** stay stationary" means that θ_{curr} is not the global maximum for $Q(\theta, \theta_{\text{curr}})$ and the impending M-step would update $\theta = \arg_\theta \max Q(\theta, \theta_{\text{curr}})$. This implies that there exists some point θ' in an arbitrarily small ϵ-neighborhood around θ_{curr} which satisfies $\mathcal{Q}(\theta', \theta_{\text{curr}}) \geq \mathcal{Q}(\theta_{\text{curr}}, \theta_{\text{curr}})$.

The existence of θ' is not general, but we can choose it to be arbitrarily close to θ_{curr}. Since θ_{curr} is a local maximum for $\ell(\theta; d)$ and we can get arbitrarily close, we can presume $\ell(\theta_{\text{curr}}; d) > \ell(\theta'; d)$.

Next, recall that the ELBO is tight at the current parameters, so $\ell(\theta_{\text{curr}}; d) = Q(\theta_{\text{curr}}, \theta_{\text{curr}})$.

Finally, all of these observations link into a syllogism:

$$\mathcal{Q}(\theta', \theta_{\text{curr}}) \geq \mathcal{Q}(\theta_{\text{curr}}, \theta_{\text{curr}}) = \ell(\theta_{\text{curr}}; d) > l(\theta'; d)$$

$$\mathcal{Q}(\theta', \theta_{\text{curr}}) > \ell(\theta'; d)$$

however, we know Q to be an ELBO for any input

$$\mathcal{Q}(\theta', \theta_{\text{curr}}) \overset{!}{\leq} \ell(\theta'; d)$$

Thus, a contraction is shown, so the assumption was false, and the implication "if the current parameters constitute a locally maximum likelihood for the GMM, the EM process will stay stationary" is proven true. □

If the EM algorithm ascends to a local max with the surrogate bound, it will stay there, but we haven't covered the if. Convergence is guaranteed, since likelihood is guaranteed to monotonically increase at each iteration.